AOPDet: Automatic Organized Points Detector for Precisely Localizing Objects in Aerial Imagery

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***Abstract*— With the development of deep convolutional neural networks, detecting rotating objects in remote-sensing images is of great significance in various fields. Existing rotating object detectors most suffer the problem of ambiguous supervision caused by inappropriate rotating object representations. This problem may result in fuzzy object localization and further lead to misclassification. In this article, we propose an Automatic Organized Points Detector (AOPDet), which derives precise localization results by applying a novel rotating object representation called nonsequential corners representation. To achieve the proposed representation, an Automatic Organization Mechanism (AOM) technique is designed to guide the model to organize points to object corners automatically. An Automatic-Organized-Points-specific (AOP-specific) head structure is also designed and equipped in the model to better focus on the rotating object detection task. On public aerial datasets, experiments show that the AOPDet achieves 17.0 mAP higher than the compared baseline model, reaching the state-of-the-art (SOTA) level. Detailed ablation experiments and error analysis strongly reveal the effectiveness of the proposed model.**

***Index Terms*— Ambiguous supervision, deep convolution neural network, object detection, remote sensing, rotating object representation.**

I. Introduction

OBJECT detection in remote-sensing images has reached substantial progress with the development of convolutional neural networks (CNNs) [1]–[11]. In contrast with the horizontal perspective view in a natural scene, aerial images are taken from the birds' eye view. As a result, objects are arbitrarily oriented and densely packed, which is more suitable for rotating object detection task [12]–[17].

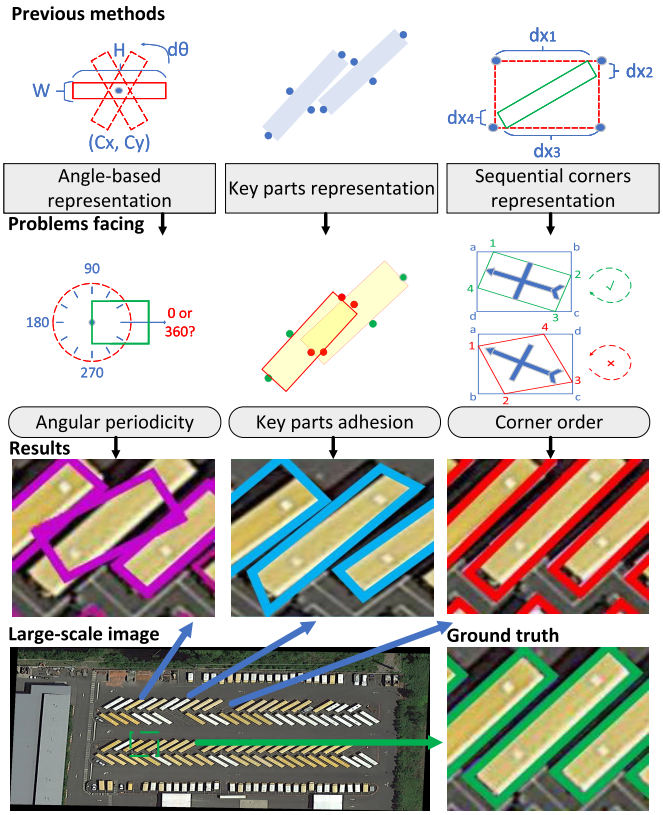

Fig. 1. Existing rotating object representations and their causes of the ambiguous supervision. Under the above circumstances, fuzzy detection results may be derived comparing with the GTs.

Three mainstream methods are aiming at this task, and they can be distinguished by the kind of rotating object representation. They are known as the angle-based representation, key parts representation, and the sequential corners representation.

The mainstream angle-based representation appends angle to rotate the axis-aligned box in angular range for matching the ground truth (GT) [18]–[21]. However, the chosen angle may be diverse due to some extreme situations. There are two cases causing the above issues. First, because of the angular periodicity, both minimum and maximum degrees are available when an object angle reaches the boundary of the angular range (see Fig. 1). Second, if representing a symmetrical

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object (common in remote-sensing images), its head and tail are indistinguishable, not mentioning angle definition. Overall, applying multiple angle choices to supervise the model (ambiguous supervision) is contradictory. It means that the model may not learn a unique rotating object representation through this angle-based representation, leading to indistinct localization results [13], [22]–[24]. Especially, a tiny angle jitter may cause huge Intersection over Union (IoU) change in rotating object detection task when detecting extreme aspect ratio objects (vehicles, ships, harbors, etc.) [25].

Key parts representation has been gradually applied to the rotating object detection task [26], [27]. It first defines key parts to represent a rotating object. Then the model predicts these instance-agnostic key parts with a pixel-wise segmentation method. Finally, post-processing is necessary to merge them to each rotating object. In some cases, the key parts may stick to each other, which we call the adhesion problem (see Fig. 1), causing ambiguous supervision and further leads to imprecise boxes. Also, to merge a large number of key parts requires heavy computational cost, which is not favorable.

Compared to the above methods, the sequential corners representation appears to be more flexible and intuitive. Although representing rotating objects with sequential corners, it has no segmentation adhesion problem and does not need the post-process of grouping. The reason is that in each feature map bin, instead of predicting instance-agnostic key parts, the regression method directly predicts a sequence of each object corner offsets. The sequential corners representation constrains the object corners sequence with a fixed pattern [25], [28], [29]. As a result, it may cause the corner order issue (see Fig. 1) for defining the corner sequence [25], causing unreasonable localization results.

After exploring the above rotating object representations, we find out that most of them suffer the problem of ambiguous supervision. This ambiguous supervision problem may hinder the model from learning precise localization and even affect the quality of classification features, leading to misclassification. In addition, the problem cannot be eliminated, for they are inherent in their representation principle (angular periodicity, key parts adhesion, and corner order definition).

To eliminate the influence of the ambiguous supervision on the model fundamentally, we design a more suitable representation strategy aiming for the rotating object, which is called the nonsequential corners representation strategy. We completely remove the corner sequence definition and let the model automatically organize the matching between the predicted points and GT through our proposed AOM. Benefit from AOM, each predicted point can be regressed to its target corner in a more reasonable manner. What is more, it liberates the limitation on the predicted point numbers, and the model obtains a better performance with the points grow. A simple yet effective conversion is applied to derive the rotating box from the predicted redundant point set. In general, our proposed method only introduces a few additional learnable variables, requiring ignorable computational cost, in exchange for more precise localization results.

Our contributions and innovations are summarized as follows: 1) We propose an Automatic Organized Points Detector (AOPDet) (see Fig. 2), which solves the ambiguous supervision problem that appears in the existing methods on densely arbitrarily oriented arranged objects in remote-sensing images. As a result, our model derives more precise localization results compared with other methods. 2) We propose a nonsequential points representation strategy, and furthermore achieve it by designing an AOM technique, which is the key to avoiding the ambiguous supervision problem fundamentally. 3) An Automatic Organized Points (AOP)-specific head structure is designed and equipped in AOPDet to generate the precise points localization, for better adaptive to the rotating object detection task.

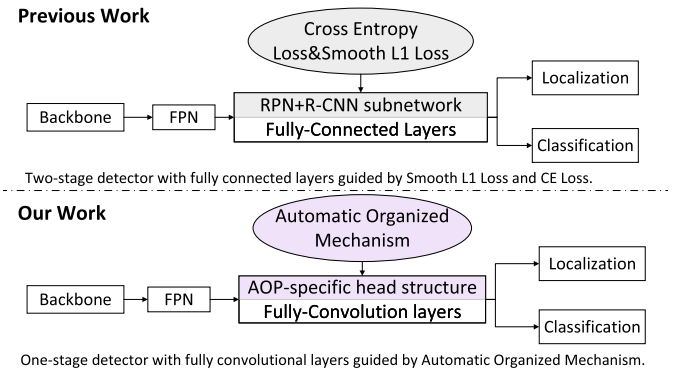

Fig. 2. Comparison of the previous work and our work. In the previous work, the detectors usually adopt Region Proposal Network (RPN) and fully connected layers, which is guided by the traditional Smooth L1 Loss and Cross-Entropy Loss. In our work, the detector removes the RPNs and adopts the Automatic-Organized-Points-specific (AOP-specific) head structure, which is guided by the proposed Automatic Organized Mechanism (AOM).

Our AOPDet outperforms other state-of-the-art (SOTA) methods on the classic remote-sensing datasets for the rotating object detection task. Experimentally, complete ablation studies are conducted to prove the effectiveness of the proposed representation strategy and techniques.

The rest of this article is organized as follows: In Section II, the related works are described. We then introduce our method in Section III. The results and analysis of the experiments are presented in Section IV. At last, we summarize our work in Section V.

II. Related Works

To better explain the details and their variants of the existing representation methods, we organize the following sections.

A. Rotating Box and Angle-Based Representation for Rotating Objects

At present, angle-based representation is the mainstream method commonly applied in the remote-sensing rotating object detection task. It introduces five parameters $(d_{cx}, d_{cy}, d_h, d_w, d_\theta)$, which refer to the offsets of midpoint, length, width, and angle of a rotating box. Aiming to match the arbitrary rotating targets (e.g., vehicle, ship, plane) in remote-sensing images, early works [23], [30]–[32] attach different angles to the preset anchor boxes in RPNs [20]. It truly derives high-quality proposals, but introduces large parameters and

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requires extra heavy computational cost as well. Apart from it, imprecise proposals predicted from RPNs may affect the classification results, which is called the feature misalignment problem [24]. rotation aware networks (RANet) [32] and Region of Interest (RoI) transformer [24] alleviate the above problem by designing rotating RoIs to warp the rotating feature from proposals. For better detecting small and cluttered rotating objects (e.g., small-vehicles (SVs), swimming pool, ships), detector for small, cluttered and rotated objects (SCRDet) [33] applies the jointly supervised pixel attention and channel attention to extract the object feature. In order to avoid the effects of complex background information appeared in remote-sensing images, context-aware detection (CAD)-Net [34] learns correlations between global-and-local object contexts and employs spatial and scale-aware attention to focus on the real targets.

B. Key Parts Representation and Segmentation Method for Rotating Objects

Apart from the angle-based representation, key parts can represent the remote-sensing rotating object more directly. One way to guide the model output object key parts is by applying the segmentation method. In natural horizontal object detection task, CornerNet [35], ExtremeNet [36], and CenterNet [37] extract instance-agnostic vertices and center points of the axis-aligned boxes and group them into objects. Nevertheless, it does not work in the remote-sensing rotating object detection task. X-LineNet [29] and DiamondNet [27] redefine the key parts of the rotating objects and design dedicated yet effective grouping methods to cluster key parts into plane or ship, respectively. Apparently, it is hard to generalize the class-specific key parts definition and grouping method to a class-agnostic level, which limits the development of it. Besides, there exist two weaknesses. First in aerial images, objects may be densely arranged, causing key parts adhesion problem and leads to the localization error. Second, if the model outputs a large number of key parts, it will be a great burden to group them into instances.

C. Sequential Corners Representation and Regression Method for Rotating Objects

In the remote-sensing rotating object detection task, sequential corners representation with the regression method is another effective way. The model regresses the sequential corners offsets of each rotating object, and by adding offsets to its corresponding bin position (from output feature map), the rotating box can be obtained. As each bin represents one single object's whole corner offsets, the regression method avoids the grouping process. In the early rotating object detection task, Rotational Region CNN(R^2CNN) [38] uses a pair of adjacent coordinates and the length of the other side (x_1, y_1, x_2, y_2, h) of the target to represent a rotating object. This representation strategy is highly dependent on the directionality of the target, while there are objects with less obvious directionality in aerial images. Gliding Vertix [25] defines the rotating box $(x, y, w, h, \alpha_1, \alpha_2, \alpha_3, \alpha_4, r)$ by regressing the offsets on the four vertices of a horizontal bounding box (HBB) sequentially. It is more suitable for remote-sensing objects compared with the angle-based representation, deriving comparative precise localization results. But it also introduces the corner order confusion issue and may lead to imprecise localization results. Rotation sensitivity detector (RSDet) [28] adopts a cross-product-based algorithm to derive the sequence of object corners, which artificially define the corner order and ease the problem.

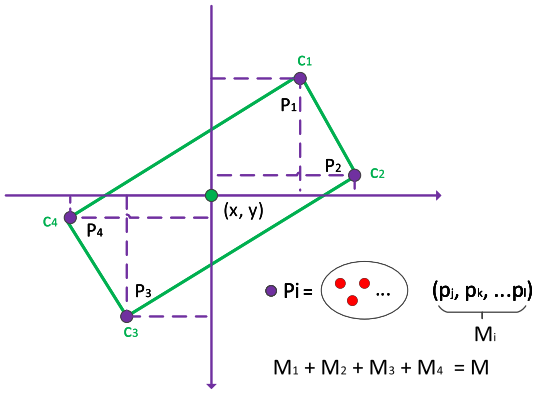

Fig. 3. In our method, the rotating object is represented as a set of nonsequential points (purple). Each corner (c_1–c_4) belongs to GT matches at least one predicted point (red), under the supervision of AOM. The number of the point set can be arbitrary as long as no less than that of the object corners.

III. Methodology

The following content will be organized into four sections. They are arranged as the problem setup, AOM, AOP-specific head structure, and loss function & post-process.

A. Problem Setup

Given a rotating object R, let $(x_i, y_i), i \in \{1, 2, \ldots, N\}$ denote the four corners of a rotating object, so the GT of a rotating object can be expressed as

$$R_{\text{gt}} = \{(x_i, y_i)\}_{i=1}^{N=4}. \tag{1}$$

Previous methods directly define the corners sequence in a fixed pattern, which causes confusion issues inevitably. Moreover, this strict matching pattern constrains the predicted point numbers and guides the model to learn quadrilateral representation rather than a rectangle. The above problems may result in imprecise localization results, thus affect the model performance. Opposite to fix the sequential corners, we define a set of points in a nonsequential way. Fig. 3 shows a visual illustration of our proposed representation. Let $p_j = (x + \Delta x_j, y + \Delta y_j), j \in \{1, 2, \ldots, M\}$ denote the point offsets predicted by our model localized in point p, so the proposed nonsequential corners representation strategy can be described as follows:

$$R_p = \varphi\left(\{p_j\}_{j=1}^{M \geq N}\right) \tag{2}$$

$\varphi(.)$ represents a mapping rule for extracting the corners of a point set, whose output dimension is consistent with that of R_{gt}. Besides, we set $M \geq N$ in order to ensure that our model can at least derive the corners distribution of a rectangle.

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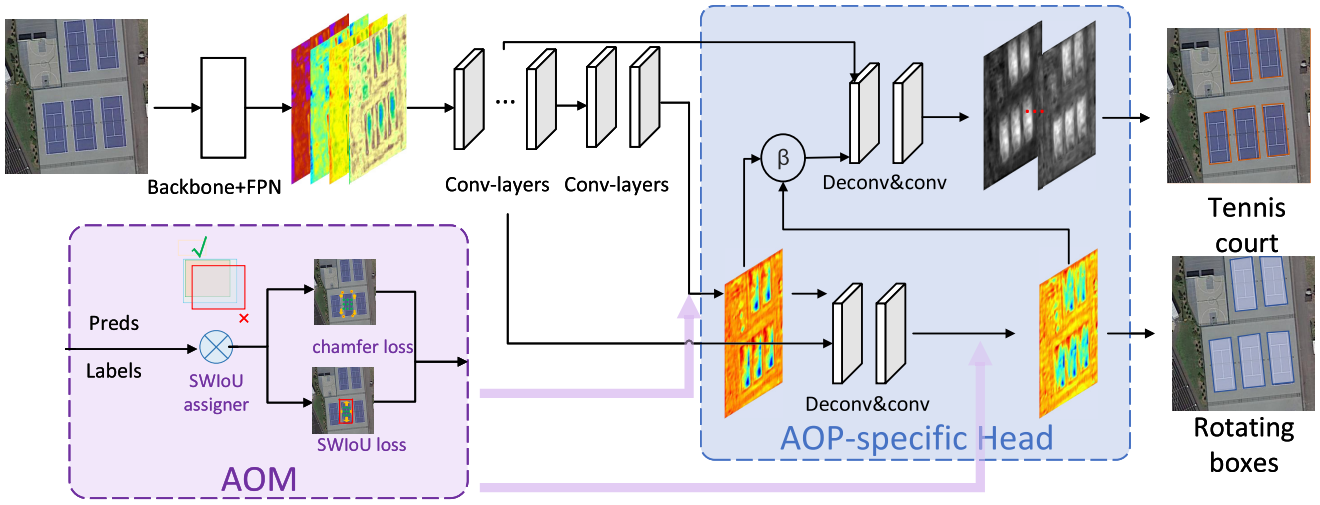

Fig. 4. Training pipeline of the proposed AOPDet applying AOM and AOP-specific head for detecting rotating objects. Images are fed into backbone plus Feature Pyramid Network (FPN) structure and then go through the AOP-specific head structure to extract the localization and classification features of the objects. AOM is equipped into the head structure to guide the model to predict precise points automatically.

We have tried different values of M in the ablation experiment. Finally, $M = 9$ is chosen for the purpose of balancing the computational cost and performance, which further proves our conjecture.

Applying the above representation strategy, we introduce the architecture of the proposed AOPDet next. As Fig. 4 shows, the network architecture is a one-stage anchor-free detector. Apart from the same backbone and FPN [4] structure as RetinaNet [39], we design a dedicated, yet effective head structure to output the redundant predicted points offsets. In general, the image is input to the backbone and FPN structure to extract multiscale features with continuous down-sampled ratios. Then the head structure slides on each feature map from different pyramid levels and outputs the corresponding localization and classification results simultaneously. Compared with two-stage detectors, the inference process of the proposed network architecture is simple yet swift.

B. Automatic Organized Mechanism (AOM)

We introduce the AOM technique to guide the model to organize the predicted points approach the object corners automatically. It consists of two parts: Chamfer loss and Self Weighted Intersection over Union (SWIoU). They are described as follows.

1) Chamfer loss: The existing Smooth L1 loss calculates the Smooth L1 distance between pairs of the points, which requires predefining the matching relationship between corners and the predicted points. However, artificially defining the order of the corners is the main factor leading to ambiguous supervision during training. In order to completely solve this problem, we introduce Chamfer loss [40]–[42] to replace the existing Smooth L1 loss. Chamfer loss calculates the Chamfer distance (CD) between the regressed point sets and the target point sets and then takes the logarithm of the distance as a loss. Model regression aims to reduce the distribution difference between the predicted point set and the GT corner set. The formula is as follows:

$$\mathrm{CD}(S,\hat{S}) = \frac{1}{2N}\sum_{i=1}^{N}\min_{j}||(x_j,y_j)-(\hat{x}_i,\hat{y}_i)||_2 + \frac{1}{2M}\sum_{j=1}^{M}\min_{i}||(x_j,y_j)-(\hat{x}_i,\hat{y}_i)||_2. \quad (3)$$

Supposing we have two sets of points S and $\hat{S}$, and the number of points in these two sets can be different. (Here S set has N points representing a rotating bounding box (RBB) and $\hat{S}$ has M points for regression.) We first establish a mapping relationship between these two sets of points. The mapping function's value is the sum distance from each point to the point belonging to another set of points with the shortest $L2$ distance. Then average both mapping function values of the two sets of points separately and finally add the two sets of points' calculation results. In order to stabilize the mapping result, we finally get the formula for Chamfer loss (L_{cd})

$$L_{\mathrm{cd}} = \log_2(\mathrm{CD}(S,\hat{S})+1). \quad (4)$$

Paying attention to the corner distribution problem discussed in the next part, we also introduce SWIoU into the loss calculation to constrain the corners distribution to obtain a more accurate RBB.

2) Self Weighted Intersection Over Union: As shown in Fig. 5, the point set obtained by the traditional IoU is evenly distributed around the corner points of an object, resulting in its rectangular outline often larger than the practical RBB of it. And this problem is more serious in the rotating object detection task. In order to solve the above problem, we optimize the Intersection over Union (IoU) calculation rules and apply it to the sample assignment and the loss function. The traditional IoU calculation formula is as follows:

$$\mathrm{IoU} = \frac{A_{\mathrm{pred}} \cap A_{\mathrm{gt}}}{A_{\mathrm{pred}} \cup A_{\mathrm{gt}}} = \frac{I}{A'_{\mathrm{pred}} + I + A'_{\mathrm{gt}}} \quad (5)$$

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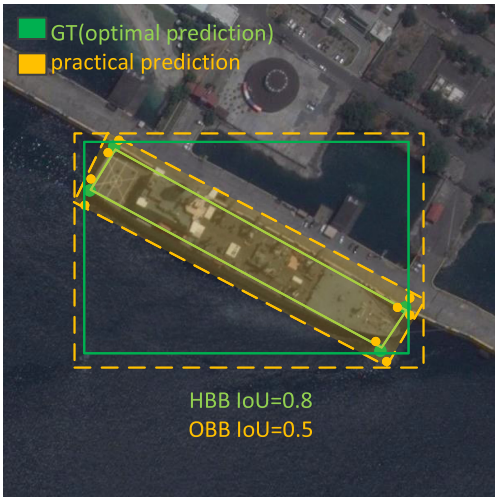

Fig. 5. Green points represent the optimal result of model convergence (consistent with the GT). In practice, the model often fails to reach the ideal state and usually converges to the orange points, resulting in its minimum rectangle larger than the GT. This situation brings greater IoU error in the RBB, compared with the HBB.

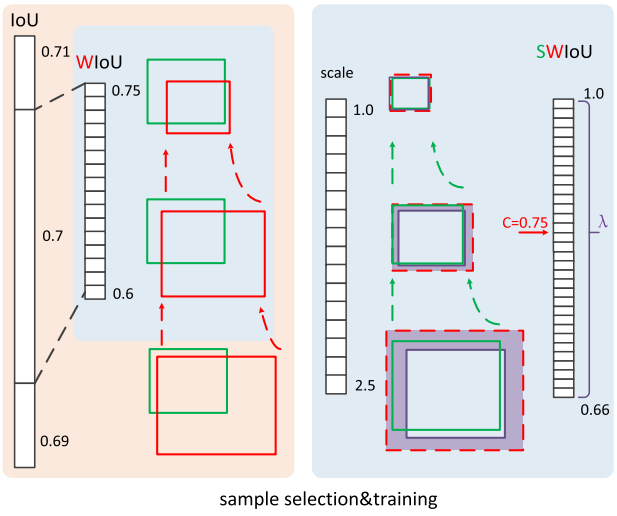

Fig. 6. (Left) When IoU fixes, WIoU has a larger range of change, corresponding to the scale of the prediction box (red). (Right) When the scale of the GT (green) changes, compared with fixed WIoU factor (red), SWIoU (purple) can dynamically adjust the weight factor λ, leading to the prediction box closer to the GT (the purple area indicates the change of the prediction box caused by the weight factor).

A_{pred} and A_{gt} represent the area of the predicted box and the GT box. I represents the intersection between A_{pred} and A_{gt}. A'_{pred} and A'_{gt} represent the area of the predicted box and the GT box minus I, respectively. We first introduce a constant factor C to weight the two parts separately for guiding the point set to converge inside the GT area. So a weighted IoU formula is expressed as follows:

$$\text{WIoU} = \frac{I}{(2-C)A'_{\text{pred}} + I + C \cdot A'_{\text{gt}}}. \tag{6}$$

Theoretically, $C \in [0, 2]$ and when C equal to 1, WIoU degenerates into IoU. If $C = 0$, it means that we only impose penalties on points that fall outside the GT, which leads to $A_{\text{pred}} \subseteq A_{\text{gt}}$. It can be seen that the ideal value of C should be in $[0, 1)$. Fig. 6 (left) shows that WIoU ($C \in [0, 1)$) can select samples with relatively smaller prediction boxes while their IoUs are almost the same.

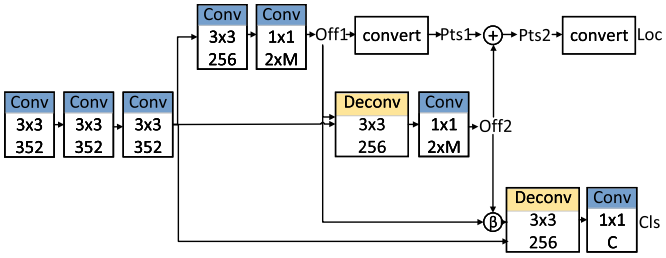

Fig. 7. Head architecture of AOPDet. C represents the number of categories to be predicted. And β is a learnable weighting factor.

At the same time, we take into account that this kind of weighting has different constraints on objects of various scales, as Fig. 6 (right) shows. In order to get the best distribution constraint on all scales of objects, we replace the factor C with a variable factor λ and make it negatively correlated with the object scale. The improved Self Weighted IoU formula is as follows:

$$\text{SWIoU} = \frac{I}{(2-\lambda)A'_{\text{pred}} + I + \lambda \cdot A'_{\text{gt}}}. \tag{7}$$

The calculation of λ is as follows:

$$\lambda = \frac{\mu}{\log_2 \sqrt{A_{\text{gt}}}} \tag{8}$$

where $\mu = 7$, we refer to the average scale of the highest accurate categories in the IoU group from the ablation experiments and bring it into formula (8) to derive the reference value of μ ($\lambda = 1$).

In general, SWIoU has solved two problems. One is to strengthen IoU's ability to constrain the point set and try to prevent the point set from falling outside the GT, as Fig. 6 (left) demonstrates. The other is to adaptively adjust the constraints for objects of various scales, aiming to generate the best RBB, as illustrated in Fig. 6 (right).

In the experimental session, we design the comparison of the results of WIoU under different values of C and SWIoU on the a large-scale dataset for object detection in aerial images (DOTA) dataset. The experimental results show that our proposed SWIoU has the best overall performance on objects of different scales.

C. AOP-Specific Head Structure

For better application to the rotating object detection task, we design the head structure (see Fig. 7) equipped with deformable convolution and Adaptive Feature Selection (AFS) module aiming for precise localization and classification, respectively.

1) Deformable Convolution for Localization: The standard convolution has a fixed sampling grid at each position of the feature maps. As a result, there exists misalignment between the axis-aligned and size-fixed sampling grid and the rotating box. We introduce deformable convolution to operate rotation-pose-aware feature extraction to ensure the alignment between the rotating box and the sampling grid. Specifically, the

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deformable convolution layer takes the rotating box as input and estimates the sampling grid to cover the object layout.

For the head structure design, three 352-d 3×3 Conv layers are first applied to extract the available features. Followed by two branches, they are aiming at localization and classification, respectively. The localization branch adds the standard 256-d 3×3 Conv layer output to the deformable 256-d 3×3 Conv layer output to further derive precise predicted points. While the classification layer applies a deformable 256-d 3×3 Conv layer, which shares the same input features with that of the localization branch.

2) AFS for Classification: Precise classification requires typical features that better represent an object. However, typical features are not easily defined and diverse on different objects. We design the AFS module to guide the head structure to select the typical object features adaptively. This module is achieved by selecting the receptive field of the classification def-Conv. The input offset field, which guides the classification def-Conv, is adaptively selected with a learnable parameter β. The process is described by the formula as follows:

$$\mathrm{Off}_{\mathrm{cls}} = \beta \cdot \mathrm{Off}_1 + (1-\beta)\mathrm{Off}_2 \tag{9}$$

where Off_1 and Off_2 are the offset outputs from localization std-Conv and def-Conv, respectively. At first, β is initiated to 1, and the learned value ranges in $(0, 1)$.

D. Loss Function and the Post-Process

Taking into account the classification loss, our final loss is as follows:

$$L = \gamma_1(L_{\mathrm{cd1}} + L_{\mathrm{swiou1}}) + \gamma_2(L_{\mathrm{cd2}} + L_{\mathrm{swiou2}}) + \gamma_3 L_{\mathrm{cls}}. \tag{10}$$

Among them, γ_1, γ_2, and γ_3 are the weighting factors. We set $\gamma_1 = 0.5$, $\gamma_2 = 2$, and $\gamma_3 = 1$ during training. Focal loss [39] is chosen as the classification loss, and the parameters in focal loss are consistent with RetinaNet [39].

The post-processing process has one more step from points to box, which is swift and parameter-free. We achieve this process through the function library provided by OpenCV.

We use rotated Nonmaximum Suppression (NMS) to suppress bad boxes and get the final high-quality rotating bounding boxes.

IV. Experiments

A. Datasets

For experiments, three classic and representative datasets are chosen for rotating object detection in aerial images. They are DOTA [43], HRSC2016 [44], and the University of Chinese Academy of Sciences (UCASs)-aerial object detection (AOD) [45].

1) *DOTA [43].* Known as the largest AOD dataset annotated with the RBB, DOTA contains 2806 large size images ranging from 800×800 to 4000×4000 pixels, with resolutions ranging from 0.75 to 2.44 m. There are 188 212 instances from 15 categories varying greatly in scale, orientation, and aspect ratio. We strictly follow the dataset process and data augmentation operated in [24] and get the same 37 373 patches for training and testing.
2) *HRSC2016 [44].* HRSC2016 is an RBB dataset with an average aspect ratio of 5:1 that contains more than 20 ship categories. The image sizes range from 300×300 to 1500×900 pixels, with resolutions ranging from 0.4 to 2 m. We also follow the same processing method of the HRSC2016 dataset in [24], including dataset division, resolution adjustment, and data augmentation.
3) *UCAS-AOD [45].* UCAS-AOD annotates 1000 images for the plane category and 510 images for the small vehicle category in rotating and horizontal bounding boxes. Its image size has about 659×1280 pixels, which come from Google Earth. In our experiment, we divide two datasets into training and test set by 8:2 randomly. Only RBB annotation is used in our experiment. Furthermore, horizontal flipping is added for data augmentation. Before training, all the images are resized to $(800, 1333)$.

B. Training and Testing Details

Our experiments are performed on single *Nvidia Tesla P100 GPU* with 16 GB memory with *PyTorch 1.3* [46]. A stochastic gradient descent (SGD) optimizer with a learning rate equals to 0.01 is applied. We have utilized 12 epochs (for ablation) and 24 epochs (for comparison) training strategies and both decrease the learning rates to 10% at two-thirds and nine-tenths of the total epochs. The weight decay and momentum are set to 10^{-4} and 0.9, respectively. The ImageNet [47] pretrained model is used for initialization. We use the sliding window method to solve the problem of large remote-sensing image scales. During the training and testing process, the raw images are cropped with a window size of 1024×1024 and are sent to detectors as patches.

C. Evaluation Metrics

We adopt the average precision (AP) and mean average precision (mAP) to evaluate the performance of the detection results. The RBB that overlaps with the GT will be considered as a candidate. Based on IoU and the classification result, evaluation can be done for whether it is correctly detected [43], [48], [49].

1) IoU: We calculate the ratio of the intersection area with the merged area from two rotating boxes as IoU. We then define the predicted box with the GT IoU greater than the predefined threshold as true positive (TP); otherwise, it is defined as false positive (FP). If a GT does not match any prediction box, it is defined as a false negative (FN).

2) AP & mAP: The precision and recall can be calculated by

$$\mathrm{precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{11}$$

$$\mathrm{recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{12}$$

As the predefined score threshold changes, the precision and recall will also change dynamically. We select the precision and recall when the score threshold equals to 0.5 [48]. After drawing the precision-recall (PR) curve, we calculate the area between the curve and the coordinate axis and view as the AP. mAP is the average value of all classes APs.

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TABLE I

SOTA COMPARISONS ON DOTA. ALL MODELS USE THE SAME BACKBONE RESNET-101 [3]. "PAR." MEANS PARAMETERS (Mb). "-R" MEANS VERSION FOR ROTATING OBJECT DETECTION TASK

Models	Plane	BD	Bridge	GTF	SV	LV	ship	TC	BC	ST	SBF	RA	Harbor	SP	HC	Par.	mAP
Two-stage																	
R-DFPN[50]	80.9	65.8	33.8	58.9	55.8	50.9	54.8	90.3	66.3	68.7	48.7	51.8	55.1	51.3	35.9		57.9
R^2CNN[38]	88.5	71.2	31.7	59.3	51.9	56.2	57.3	90.8	72.8	67.4	56.7	52.8	53.1	51.9	53.6		60.7
RRPN[23]	80.9	65.8	35.3	67.4	59.9	50.9	55.8	90.7	66.9	72.4	55.1	52.2	55.1	53.4	48.2		61.0
Yang et al.[31]	81.3	71.4	36.5	67.4	61.2	50.9	56.6	90.7	68.1	72.4	55.1	55.6	62.4	53.4	51.5		62.3
FR-R[20]	88.0	77.0	36.7	72.5	70.2	61.8	75.8	90.1	73.8	85.0	56.6	62.6	53.3	59.5	41.9	41.1	67.0
ICN[51]	81.4	74.3	47.7	70.3	64.9	67.8	70.0	90.8	79.1	78.2	53.6	62.9	67.0	64.2	50.2		68.2
RoITrans.[24]	88.6	78.5	43.4	75.9	68.8	73.7	83.6	90.7	77.3	81.5	58.4	53.5	62.8	58.9	47.7	54.8	69.6
CAD-Net[34]	87.8	82.4	49.4	73.5	71.1	63.5	76.6	**90.9**	79.2	73.3	48.4	60.9	62.0	67.0	62.2		69.9
SCRDet[33]	90.0	80.7	52.1	68.4	68.4	60.3	72.4	**90.9**	**87.9**	**86.9**	**65.0**	66.7	66.3	68.2	65.2		72.6
FR-Est [52]	89.6	81.2	50.4	70.2	73.5	78.0	86.4	90.8	84.1	83.6	60.6	66.6	70.6	66.7	60.6		74.2
Gliding Vertix [25]	89.6	**85.0**	52.3	**77.3**	73.0	73.1	86.8	90.7	79.0	86.8	59.6	70.9	72.9	70.9	57.3		75.0
One-stage																	
SSD-R[53]	39.8	9.1	0.6	13.2	0.3	0.4	1.1	16.2	27.6	9.2	27.1	9.1	3.0	1.1	1.0		10.6
YOLOv2-R[54]	39.6	20.3	36.6	23.4	8.9	2.1	4.8	44.3	38.4	34.7	16.0	37.6	37.2	25.5	7.5		21.4
RetinaNet-R[39]	81.1	53.2	33.6	45.7	48.5	55.7	75.9	90.0	42.6	64.3	44.4	59.3	54.3	61.7	49.1	36.7	57.3
R3Det[55]	89.5	82.0	48.5	62.5	70.5	74.3	77.5	90.8	81.4	83.5	62.0	59.8	65.4	67.5	60.1		71.7
O^2-DNet[29]	89.3	83.3	50.1	72.1	71.1	75.6	78.7	**90.9**	79.9	82.9	60.2	60.0	64.6	68.9	**65.7**		72.8
S^2A-Net[56]	88.7	81.4	**54.3**	69.8	**78.0**	**80.5**	88.0	90.7	84.8	86.2	**65.0**	65.8	**76.2**	**73.4**	58.9		**76.1**
AOPDet	**90.2**	77.9	47.5	74.3	73.6	78.7	**89.5**	**90.9**	66.4	85.7	57.1	**71.9**	75.6	72.8	60.9	**36.6**	74.3

3) Error Analysis: To further inspect the distribution of the false-positive results from AOPDet, we introduce the following types from the previous works [52], [57], [58] to conduct analysis on FP results.

1) *Cor:* The true-positive sample (IoU $>$ 0.5) is treated as a correct sample.
2) *Loc:* Localization error will be decided if an FP sample has correct category but overlaps less with GT ($0.1 <$ IoU < 0.5).
3) *Oth:* Confusion with other categories will be treated if an FP sample overlaps (IoU $>$ 0.1) GT but incorrectly classified.
4) *BG:* Confusion with background will be decided if an FP does not overlap (IoU $<$ 0.1) any GT.

We focus on localization error and classification error to further check whether the proposed techniques actually make effective improvements in their child task, respectively. The above error analysis metrics will be applied in ablation experiments.

D. Comparisons With SOTA Frameworks

1) Comparisons on Categories: We compare the performance of our AOPDet with the classic angle-based representation and sequential corners representation RBB detectors on three datasets under the same experimental environment.

For the baseline detector, a modified RetinaNet-R [39] detector for the RBB is provided.

a) Experiment on DOTA: We compare the performance of AOPDet with other SOTA detectors on the DOTA dataset. Annotations of rotating bounding boxes are applied to all compared models. Noticing that other detectors [24], [33], [34], [39], [52] utilize ResNet-101 [3] with FPN [4] as backbone architectures, we instantiate our AOPDet with the same structure for fair comparison. Only the AOPDet applies the nonsequential corners representation strategy, while other detectors still apply angle-based representation or sequential corners representation.

Table I shows the results. In 15 categories of the DOTA dataset, our model reaches the SOTA level in five categories. Finally, our AOPDet achieves a mAP of 74.3%, outperforming all listed one-stage and two-stage object detectors. Competitive two-stage detectors (e.g., RoI transformer, CAD-Net, SCRDet, faster-RCNN (FR)-point-based ESTimator (EST) [24], [31], [34], [52]) apply fully connected layers to regress the predicted rotating box. Our AOPDet is operated with a fully convolutional network, which requires fewer parameters and computational cost. Compared with the baseline RetinaNet-R [39], AOPDet achieves significant boost of 17.0% mAP under the same backbone architectures except for the rotating object representation strategy. O^2D-Net achieves outstanding performance in one-stage detectors, but it employs Hourglass-104 [61] as its backbone architecture, which takes larger parameters and computational cost compared with Resnet-101 with FPN.

Figs. 8 and 10 illustrate some visualization results of our model on the DOTA dataset. We can find that our model has performed accurate localization capabilities on objects of different scales, categories, extreme aspect ratios, and even slight angle changes. Specifically, as Fig. 8 SV detection

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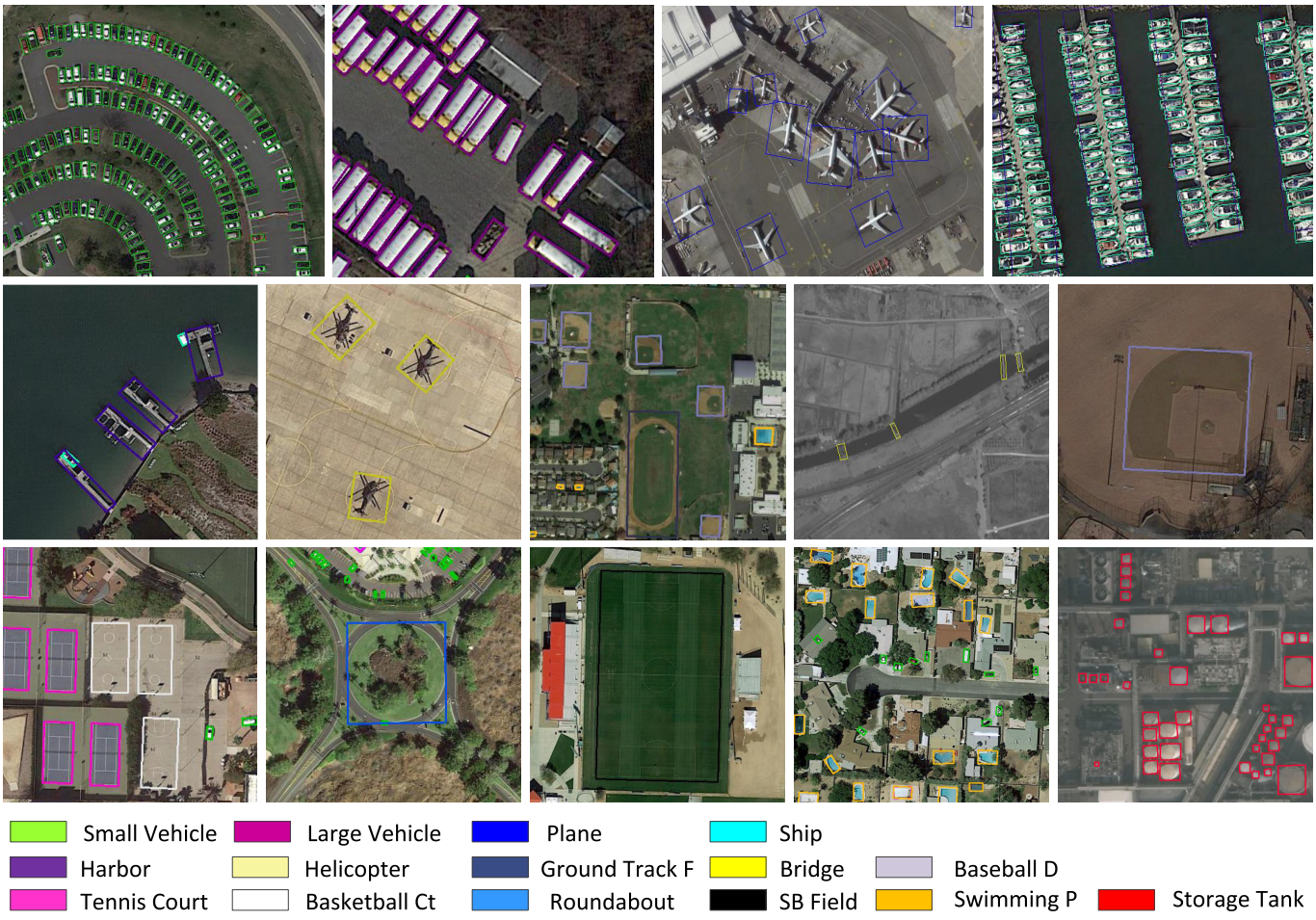

Fig. 8. Visualization of results from AOPDet on DOTA.

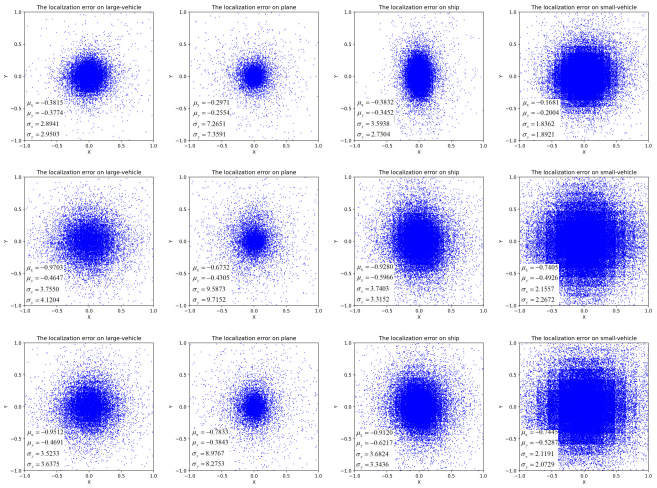

Fig. 9. (First row) Distribution of localization errors of AOPDet. (Second row) RoI transformer. (Third row) Gliding Vertex. μ_x, μ_y and σ_x, σ_y represent the means and standard deviations of the localization errors along x- and y-axes, respectively.

results show, boxes perfectly cover the SVs' body, with the smooth changes of their direction angles. Arbitrarily oriented and direction-elusive objects appeared in Fig. 8 are also well detected by the AOPDet. Fig. 10 visualize several large-scale images covering multilabels and complex background, and the detection results are precise and effective. In Section IV-D2, we further explore the angular sensitivity of AOPDet, RoI transformer [24], and Gliding Vertex [25], which applies different rotating object representations.

b) Experiment on HRSC2016: The HRSC2016 dataset covers a large number of ships with an average aspect ratio close to 5:1. Due to the arbitrariness of direction and the ship's extreme aspect ratio, most anchor-based methods need to modify the corresponding hyperparameters to adapt to the new dataset. For our AOPDet represents the rotating object in an anchor-free way, there is no need to modify any hyperparameters, showing a strong ability to adapt to such datasets.

Table II shows the performance of our model on the HRSC2016 dataset. Our AOPDet reaches a mAP of 88.5%, which exceeds all other models. Fig. 11 shows some of the visualization results. It can be seen that our model shows favorable detection capabilities on ships of variable scales, different rotation directions, and extreme aspect ratios.

c) Experiment on UCAS-AOD: In order to explore the detection performance of our model on aircraft and small vehicles, we continue to conduct experiments on the UCAS-AOD dataset. Our model has reached the SOTA level, and the

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TABLE II
COMPARISONS WITH THE SOTA MODELS ON HRSC2016

method	CP[59]	BL2[59]	RC1[59]	RC2[59]	R^2PN[30]	RRD[60]	RoI Trans.[24]	AOPDet
mAP	55.7	69.6	75.7	75.7	79.6	84.3	86.2	**88.5**

Fig. 10. Visualization results of our AOPDet on large-scale remote-sensing images of complex scenes and multiple categories from the DOTA test dataset.

visualization results are shown in Fig. 11. The detection results of the aircraft show that our model is sensitive to the angle of an object, while detection results of small vehicles reflect the strong detection ability of our model to small-scale objects.

2) Comparisons on Localization Error: In order to further explore the localization capability of our model in detecting rotating objects, we design the following comparative experiments.

We first visualize some detailed detection results from the proposed AOPDet, RoI transformer [24], and Gliding Vertix [25] (see Fig. 12). Obvious localization differences can be observed which strongly demonstrates the effectiveness of the proposed method.

For further explore the localization capability of different models, we conduct analysis on our AOPDet and other two methods. We calculate each distance between the corners of Gt and results, and z-score is applied to normalize the distances. The results are listed in Fig. 9.

We conduct comparison experiment on categories of large-vehicle, plane, ship, and SV. It demonstrates that AOPDet has relatively smaller means and standard deviations of the corners localization error, further explaining a stronger localization capability than the other two.

E. Ablation Studies

1) Localization Loss: We combine Chamfer loss and SWIoU loss as our final localization loss. In order to verify the effect of these two combined losses on corner point offsets in the process of guiding model convergence, we design the following ablation experiments. The experiments are conducted on the DOTA dataset. In order to speed the efficiency, we use

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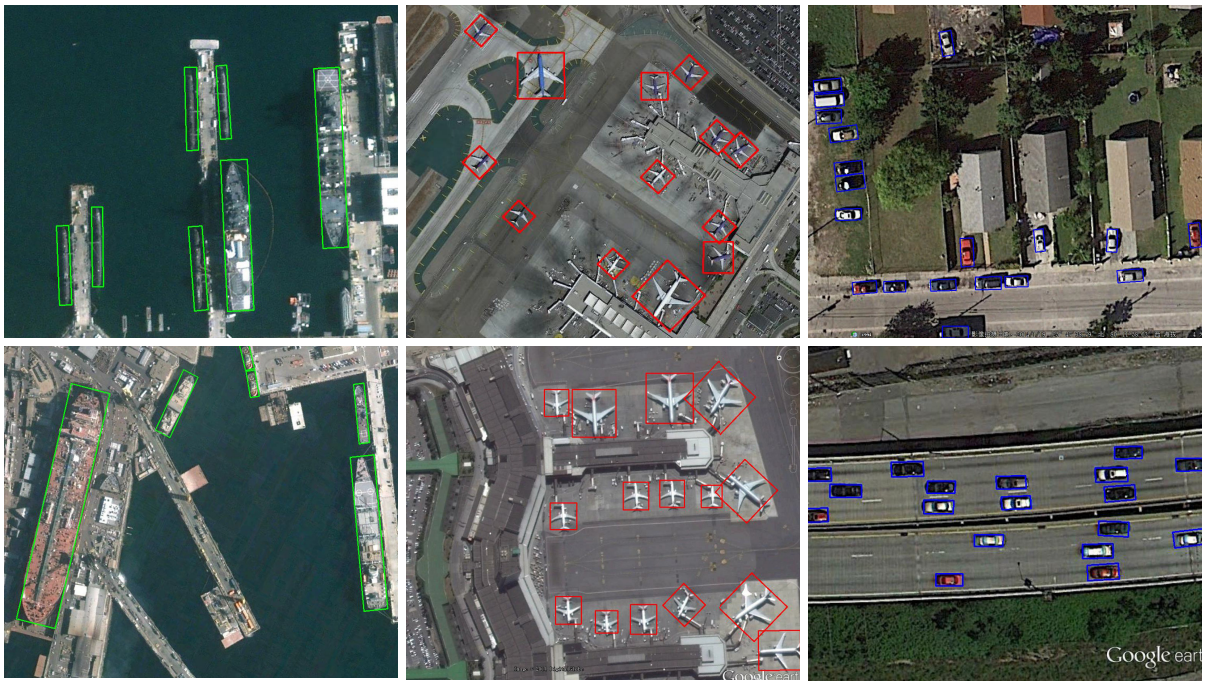

Fig. 11. Visualization results from our AOPDet on HRSC2016 and UCAS-AOD.

TABLE III
COMPARISONS WITH THE SOTA MODELS ON UCAS-AOD

models	plane	car	mAP
RRPN[23]	88.0	74.4	81.2
R2CNN[38]	89.8	78.9	84.3
R-DFPN[50]	88.9	81.3	85.1
AOPDet	**90.6**	**84.4**	**87.5**

TABLE IV
COMPARISONS OF AOPDET APPLYING TWO LOSSES OR NOT. THE EXPERIMENTS ARE CONDUCTED ON DOTA VALIDATION DATASET, AND "-" INDICATES THAT THE MODEL COULD NOT CONVERGE

Methods	SWIoU Loss	Chamfer Loss	mAP
AOPDet	✓		46.1
		✓	-
	✓	✓	67.7

Resnet-50 [3] as our backbone to train our model on the training set and test the results on the validation set without using any tricks.

We conduct comparative experiments on AOPDet. Table IV shows the performance difference in the model. Experimental results show that SWIoU loss is essential to the model's convergence, and Chamfer loss provides a more optimal distribution for the predicted points.

2) Self Weighted IoU: As mentioned in Section III-B2, Self Weighted IoU is an extended IoU calculation method that can adaptively adjust the constraints on predicted points according to the object scale. We achieve self-adaptation by setting the weighting factor λ to be negatively correlated with the scale of an object. The SWIoU technology can replace the sample assignment process and the IoU loss calculation without introducing any parameters. Based on these parts, we design the following comparative experiments to verify the rationality of SWIoU. The experiments consisting of IoU, WIoU, and SWIoU are conducted to verify the effectiveness of our proposed method.

We reorganize the objects in Table V according to the order of the objects in Fig. 14. As shown in Table V, the highest AP of each category is obtained under different values of λ. After further comparing the object size distribution in Fig. 14 and the experimental data of WIoU in Table V, it can be intuitively found that the object size and the λ value corresponding to the highest AP are negatively correlated. λ is used to measure the possibility for predicted points that fall beyond the GT. The smaller the λ, the lower the probability that the point falls outside the GT. If an object is large, λ should be decreased to strengthen this restriction. On the contrary, we will appropriately larger λ to weaken the requirements.

Based on the above analysis, we have designed a λ calculation formula that can be adaptively changed with the object scale, which we call the SWIoU. Regarding the selection of μ in formula 8, we conduct ablation experiments in Table V. The experimental results show that when μ equals to 7, the model performs best. It also demonstrates that the proposed SWIoU can calculate adaptive λ for various categories outstandingly, deriving the optimal AP of each category.

We choose the best WIoU visualization result ($\lambda = 0.5$) to compare with that of SWIoU. As shown in Fig. 13, we summarize the following problems: 1) For nonconvex objects such as airplanes, inappropriate λ constraints may

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TABLE V

APs OF ALL CATEGORIES UNDER IoU, WIoU, AND SWIoU ON DOTA VALIDATION DATASET. THE HIGHEST AP FOR EACH CATEGORY IS BOLD ON IoU, WIoU, AND SWIoU, RESPECTIVELY

IoU	λ	μ	SV	ship	ST	LV	SP	HC	bridge	RA	TC	plane	BC	BD	harbor	SBF	GTF	mAP
IoU	-	-	62.1	**85.5**	79.2	70.5	62.2	49.7	40.3	68.5	**90.4**	83.6	46.8	65.8	67.0	40.2	**60.3**	64.8
WIoU	3/2	-	**66.7**	82.5	**79.5**	68.1	**63.2**	55.5	40.9	62.3	89.1	78.6	48.5	62.6	59.5	37.9	54.4	63.3
	1/2	-	59.4	**85.2**	76.0	**69.0**	61.2	**57.9**	**42.9**	**72.3**	**89.5**	83.0	52.4	68.9	69.5	47.9	51.8	65.8
	1/4	-	54.3	82.9	78.1	68.5	59.2	56.7	42.4	68.9	86.0	86.0	**53.6**	69.1	**70.3**	**48.3**	56.5	65.4
	1/8	-	48.8	81.7	72.4	66.0	53.8	54.3	38.8	68.2	80.7	**88.2**	51.3	**69.8**	67.6	42.7	**58.0**	62.8
SWIoU	-	6	65.6	**83.6**	76.9	70.5	66.3	58.3	**41.1**	69.4	88.7	87.9	50.5	65.6	68.4	44.6	55.7	66.2
	-	7	**67.6**	83.2	**77.9**	**71.3**	**66.7**	**59.2**	40.7	**71.4**	90.4	88.1	**53.8**	**69.5**	**70.4**	**48.8**	57.7	**67.7**
	-	7.5	66.7	81.4	75.1	69.7	65.1	56.7	39.0	67.7	**90.5**	**88.9**	52.7	66.7	66.6	47.3	55.6	66.3
	-	8	63.9	79.0	72.1	68.8	66.1	57.7	37.7	68.2	88.9	87.3	51.1	67.2	65.6	46.6	**58.2**	65.2

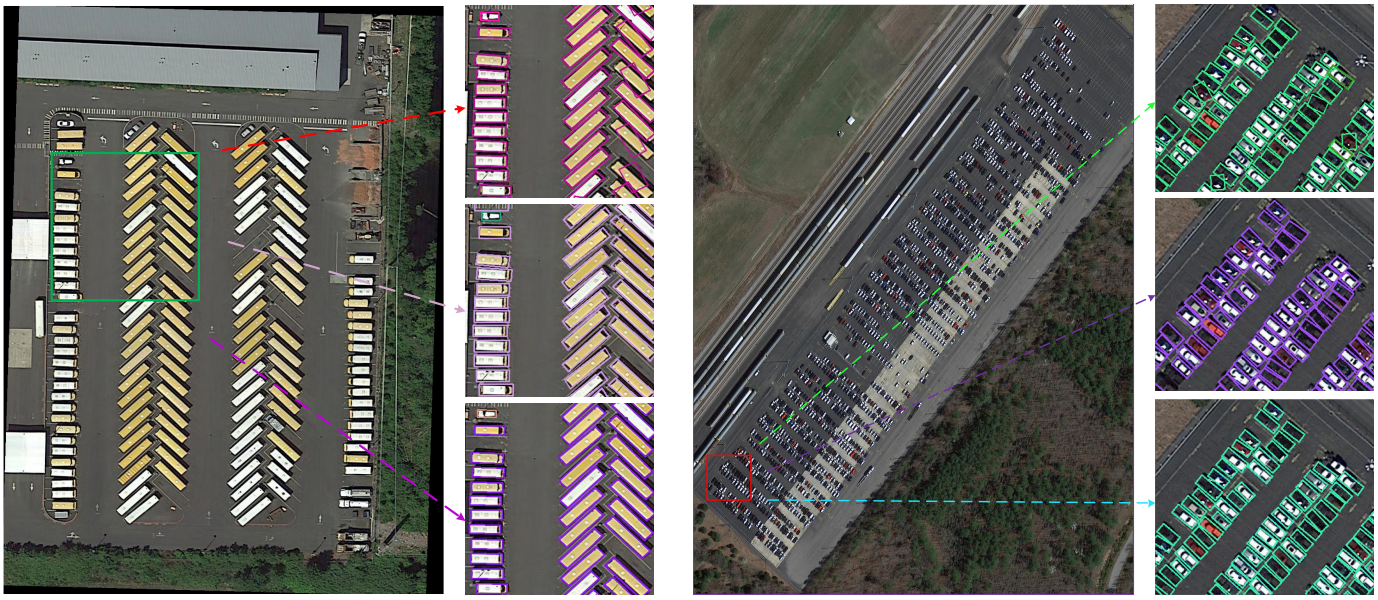

Fig. 12. Partial enlarged comparison on RoI transformer (first line), Gliding Vertix (second line), and AOPDet (third line).

result in the blurred direction of an RBB; 2) The fixed value of λ may cause unbalanced constraints on objects of different scales, leading to the scale difference between the prediction boxes and the GTs. The visualization results of SWIoU show that the above problems have been basically solved.

3) Point Set Number Selection: In Section III-A, we point out that the number of predicted point sets should be $M \geq N$. In this section, we will explore the influence of the value of M on the experimental results. Considering meeting the convolution kernel requirement, the value of M should be no more than its kernel size. Several experiments are conducted on three groups of convolutions with different kernel sizes (3×3, 4×4, 5×5). For 3×3 convolutions, we set up four sets of experiments according to the value of M from 4 to 9, with a step size of 2. For 4×4 and 5×5 convolutions, the maximum value of its kernel size is adopted. When $M = N = 4$, we additionally add the Smooth L1 loss experiment as the baseline. The experiments are carried out on the DOTA validation dataset.

We have tried several different numbers of point sets under the combination of Chamfer loss and SWIoU loss. It can be found that as the number of point sets increases, the performance of the model is also improving. When $M = 25$, we get the best result (considering maintaining the receptive field and parameters of the 5×5 convolution kernel, $M_{\max} = 25$). The experimental results further support our conjecture that in the process of fitting the corners of an object, the redundant point set can derive a better fitting effect. At the same time, we also apply Smooth L1 loss on the basis of AOPDet to regress the four-corner offsets of the RBB in the clockwise order. As shown in Fig. 15, under the same number of training steps, optimizer, and learning rate strategy, the combination of Chamfer loss and SWIoU loss is better than Smooth L1 loss in terms of loss convergence. Table VII also demonstrates that under the same amount of predicted points, Chamfer loss with SWIoU loss derives a better mAP than Smooth L1 loss. The above experimental results strongly prove the effectiveness of the proposed nonsequential corners representation strategy.

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Fig. 13. Comparison of the visualized results of corner points distribution and RBB generation under SWIoU and WIoU ($\lambda = 0.5$) among the selected objects (plane, SV, RA). Red represents WIoU, while blue represents SWIoU.

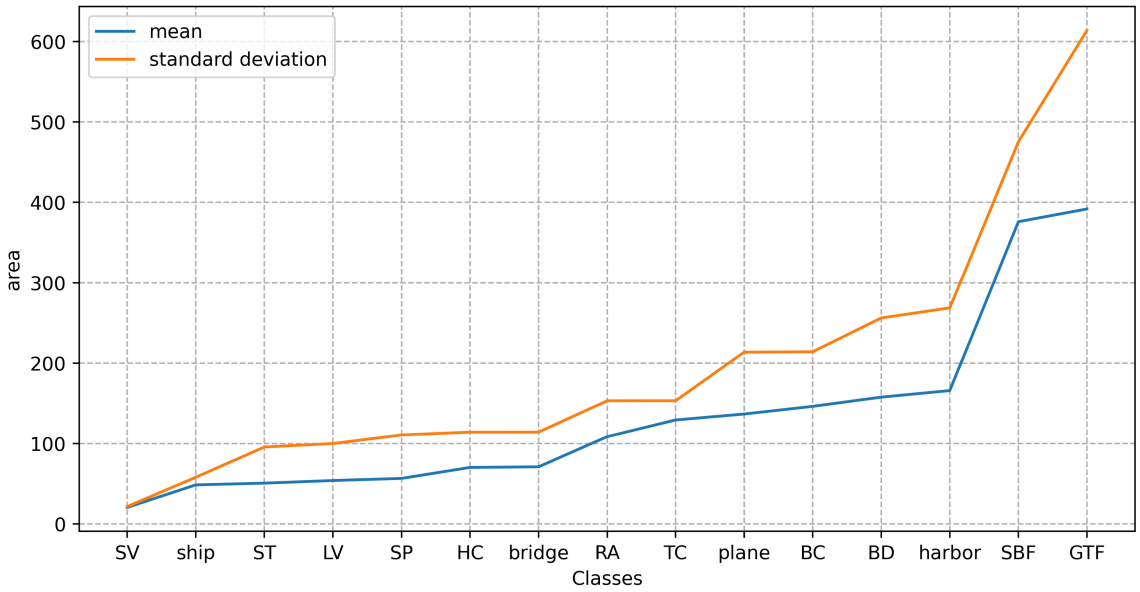

Fig. 14. Plot shows the distribution of each category instances from DOTA dataset sorted by the mean scale (area $= \sqrt{h \times w}$) from small to large (from left to right).

4) Head Architecture: To verify the effectiveness of our two improvements on the head structure, we design the following ablation experiments. Where AOP-std means only standard convolution is applied for the head structure. AOP-def means applying deformable convolution while its classification input offset field is from localization offset$_1$. AOP-def+AFS means the proposed head structure, which applies deformable convolution and AFS module.

We first analyze the results of the two groups of ablation experiments in Table VIII. In the localization branch, we compare the performance of the standard convolution, the original deformable convolution [62] and the deformable convolution with intermediate supervision (ours) on the DOTA dataset. While in the classification branch, we introduce two additional operations, namely simple addition and feature concatenation. These two sets of ablation experiments, respectively, verify

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TABLE VI

ERROR ANALYSIS ON DOTA VALIDATION DATASETS CONDUCTING HEAD STRUCTURE ABLATION EXPERIMENTS. RESNET-101 WITH FPN IS APPLIED AS BACKBONE ARCHITECTURE

Metric	Method	plane	BD	bridge	GTF	SV	LV	ship	TC	BC	ST	SBF	RA	harbor	SP	HC	mAP
Cor	AOP-std	89.8	69.3	32.8	66.1	57.6	71.9	79.9	90.8	64.0	77.8	60.6	62.3	61.9	51.9	52.1	65.9
	AOP-def	89.9	70.7	36.4	68.8	62.1	73.0	87.3	90.8	65.0	78.8	62.5	68.0	63.8	60.6	62.7	69.4
	AOP-def+AFS	90.0	72.2	40.3	68.8	67.3	73.4	88.4	90.8	65.3	79.4	62.7	69.1	71.1	65.4	64.3	71.3
Loc	AOP-std	0.3	9.2	23.6	7.3	21.6	7.6	9.8	0.0	3.4	8.3	6.5	10.3	23.6	27.7	19.2	11.9
	AOP-def	0.3	8.1	21.7	4.6	17.4	6.8	2.4	0.0	2.5	7.3	4.9	4.6	22.4	19.5	8.5	8.6
	AOP-def+AFS	0.2	7.6	19.7	4.6	15.6	6.8	2.4	0.0	2.1	7.0	6.2	3.9	21.6	18.7	8.0	8.2
Oth	AOP-std	0.1	4.0	5.4	0.0	1.2	1.2	0.3	0.0	1.9	0.6	3.1	1.8	2.9	4.3	2.1	1.8
	AOP-def	0.1	3.8	4.0	0.0	1.0	1.2	0.1	0.0	1.6	0.5	2.1	1.6	2.3	2.9	1.2	1.6
	AOP-def+AFS	0.1	2.7	2.6	0.1	0.4	0.3	0.1	0.0	1.6	0.2	1.1	2.0	1.6	0.9	0.0	0.9
BG	AOP-std	9.8	17.5	38.2	26.6	19.6	19.3	10.0	9.2	30.7	13.3	29.8	25.6	11.6	16.1	26.6	20.4
	AOP-def	9.7	17.4	37.9	26.6	19.5	19.0	10.2	9.2	30.9	13.4	30.5	25.8	11.5	17.0	27.6	20.4
	AOP-def+AFS	9.7	17.5	37.4	26.5	16.7	19.5	10.1	9.2	31.0	13.4	30.0	25.0	5.7	15.0	27.7	19.6

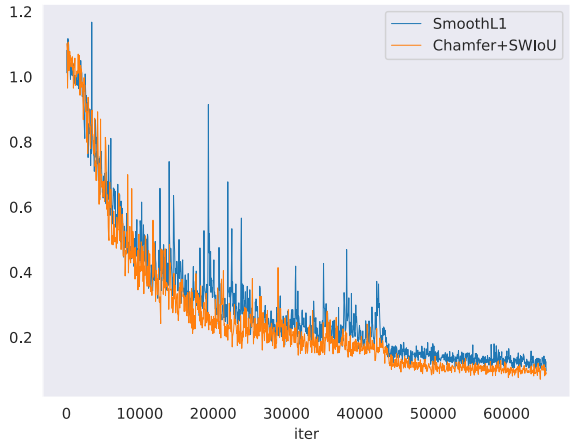

Fig. 15. Comparison of Chamfer Loss plus SWIoU Loss ($M = 9$) and Smooth L1 Loss on the total loss curve (both are normalized in a range from 0 to 1).

the effectiveness of our proposed methods. Next, we conduct a more detailed error analysis on the two subtasks of localization and classification.

As Table VI shows, AOP-def has 3.5% Cor higher than baseline AOP-std and lowers the localization error from 11.9% mAP to 8.6% mAP. Most categories reach considerable improvements on localization error with deformable convolution layers added. It verifies that adding deformable convolution layers can extract rotation-pose-aware features and successfully align the rotating box and sampling grid, deriving more precise localization results. After applying the AFS module, AOP-def+AFS reaches the improvement of 1.9% Cor compared with AOP-def. It mainly includes the 0.7% Oth and 0.8% BG, indicating that AFS provides the model with more accurate classification results.

5) Influence of Different Backbones: Compared with RetinaNet [39] as the baseline, we conduct ablation experiments on different backbones with or without FPN structure.

TABLE VII

COMPARISONS OF AOPDET APPLYING DIFFERENT POINT SET NUMBERS. THE EXPERIMENTAL RESULTS ARE CONDUCTED ON DOTA DATASET

Conv.	Num.	Loss	mAP
3x3	4	Smooth L1	64.2
3x3	4	Chamfer+SWIoU	65.4
	6		66.7
	8		67.4
	9		67.7
4x4	16		68.0
5x5	25		**68.1**

TABLE VIII

ABLATIONS OF AOP-SPECIFIC HEAD ON TWO MOTIVATIONS APPLYING DIFFERENT CONFIGURED METHODS. AOP-DEF* MEANS APPLYING STANDARD DEFORMABLE CONVOLUTION. AND $\bigoplus$ MEANS CONCATENATION OPERATION

Localization	Classification	mAP
AOP-std	-	65.9
AOP-def*	-	66.4
AOP-def	-	67.7
-	+	66.4
-	$\bigoplus$	67.6
-	AFS	67.7

Experimental results reveal that under different backbones with or without FPN, our AOPDet shows overwhelming advantages on mAP (improvement of 14% on average) stably with an ignorable extra computational cost. Especially, AOPDet with ResNet-50 without FPN still reaches the improvement of 8.8% mAP compared with RetinaNet-R with ResNet-101 FPN and requires less inference time. The above

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TABLE IX
COMPARISONS OF AOPDET AND BASELINE APPLYING DIFFERENT BACKBONES ON DOTA VALIDATION DATASET. INF TIME MEANS INFERENCE TIME (ms) PER IMAGE

Method	backbone	FPN	mAP	Inf time
RetinaNet-R	ResNet-50		51.1	65
AOPDet	ResNet-50		64.9	71
RetinaNet-R	ResNet-101		53.9	92
AOPDet	ResNet-101		68.2	96
RetinaNet-R	ResNet-101	✓	56.1	109
AOPDet	ResNet-101	✓	71.3	114

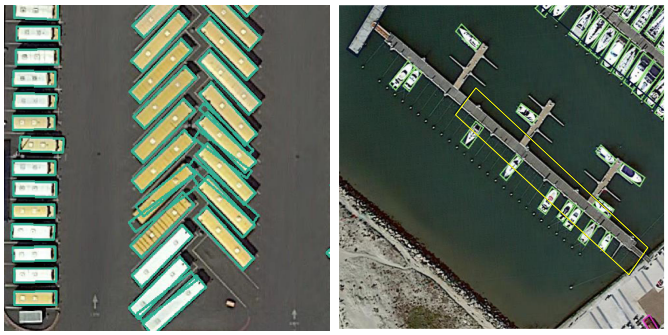

Fig. 16. Error analysis of the results from AOPDet on DOTA.

results strongly demonstrate the effectiveness and robustness of the proposed representation strategy and techniques.

6) Error Analysis: After checking the visualization results (see Fig. 16), we also find a few error detections, which can be summarized as listed. 1) Redundant false alarms may appear if objects with an extreme aspect ratio (e.g., large vehicle) arrange densely. 2) Detection results may not cover the whole object if the target has a large scale (e.g., harbor). 3) Object missing is the main error illustrated in Table VI, and it appears to be more serious in categories having large scale (e.g., ground track field, basketball court, soccer ball field).

The first factor is mainly caused by the wrong classification results, which are given by the improper classification features. The part between two objects should be recognized as background, but the AFS module may choose the object feature from the objects nearby, leading to misclassifications. The second and the third factors are probably the lack of global information when extracting the object features. The localization branch can only reach the limited boundary, so it cannot represent the whole object.

Dealing with the first problem, the classification results may be better by introducing more weights on the features from its local position when recognizing the corresponding background. The second and the third problems are hard to solve, as the same situations appear in other detectors (including RoI transformer [24], Gliding Vertix [25], etc.). Weak ability to capture long-range dependencies is the main factor. High-quality feature maps may be derived by introducing attention technology [63] and further improves the large-scale object localization results.

In general, the limitations of our method lie in the fuzzy recognition of certain objects. And the advantage lies in the precise localization of objects densely and arbitrary arranged in the remote sensing images, which benefits from the rotating objects representation method we proposed.

V. CONCLUSION

In conclusion, we propose a novel nonsequential corners representation strategy with the regression method for rotating objects to solve the ambiguous supervision problem. We design the AOM to guide the model implicitly organize the predicted points to their suitable corners. AOM consists of two parts: Chamfer loss and SWIoU, and they are employed to sample assignment and loss function. Without a fixed pattern, the predicted points can be arbitrary as long as larger than the number of object corners. We instantiate our model called AOPDet and design an AOP-specific head structure to fit the rotating objects. The experimental results show that our model can adapt well to objects of different scales, angles, and variable aspect ratios, providing more accurate localization of the rotating objects. On the DOTA, HRSC2016, and UCAS-AOD datasets, our model has reached the SOTA level. In the next work, we will focus more on improving the weakness observed in the error analysis to strengthen the classification capability of the proposed model.

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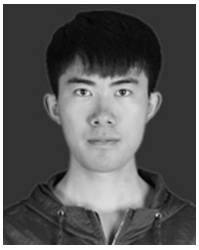

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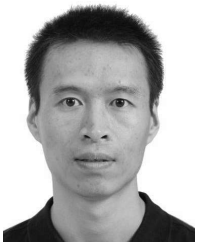

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